

Xavier Rao

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Objective

Software Engineering student seeking an internship to grow technical skills and gain hands-on industry experience. Available January to August 2027.

Education

Rochester Institute of Technology (RIT)

Bachelor of Science, Software Engineering

Expected May 2028

Projects

Notescribe, *Class Project*

Spring 2026

- Designed a decoupled, event-driven backend with SQS queues and EC2 auto-scaling workers; implemented serverless APIs with API Gateway and Lambda, secured per-user file ownership with Cognito MFA, and integrated Transcribe and Bedrock to automate lecture transcription and structured study summaries
- Collaborated in a team of five to build and deploy the cloud-native application
- Technologies: AWS (EC2, S3, SQS, DynamoDB, Transcribe, Bedrock, Lambda, Cognito), Terraform, Python, JavaScript

Most Likely To Game, *Personal Project*

May 2025 – April 2026

- Developed and deployed a real-time multiplayer party game with an imposter mechanic where one player secretly receives a different AI-generated question each round
- Technologies: React, Node.js, Socket.IO, JavaScript

CineVision, *Class Project*

Spring 2026

- Built and evaluated Neural Collaborative Filtering and SVD movie recommendation models against a baseline on the MovieLens 100K dataset, achieving measurable RMSE improvement
- Technologies: Python, PyTorch, scikit-learn, pandas

NutriApp, *Class Project*

Spring 2025

- Developed a multi-user nutritional planning app with calorie tracking, meal planning, workout logging, and automated goal transitions
- Technologies: Java, Angular, TypeScript

Work Experience

TAG Online Inc. | Clifton, NJ

Fall 2026

Intern

- Developed and maintained custom websites using WordPress, HTML, CSS, and PHP, implementing features and updates based on client requirements
- Assisted with software and systems projects, including email server management and system administration
- Researched and integrated AI tools to streamline website development workflows, improve efficiency, and automate repetitive tasks

MLC STEM Center | Bloomfield, NJ

Summer 2022/2024/2025/2026

STEM Mentor/Counselor

- Developed and delivered lesson plans on coding, physics, chemistry, and 3D design for classes of 20 students
- Designed and led a week-long curriculum introducing artificial intelligence concepts through hands-on coding projects

NanoGraf | Chicago, IL

June 2021 – July 2021

Intern

- Engineered high-performance battery cells by formulating specialized slurries and applying precision compression in a controlled glove box environment
- Conducted patent research and competitive analysis, documenting findings in detailed spreadsheets to support materials innovation

Skills

Languages: Java, Python, JavaScript, Typescript, HTML/CSS, SQL, C, PHP, Terraform

Frameworks: Angular, React.js, Node.js, Flask, Spring Boot, Express.js

Software: Visual Studio Code, IntelliJ IDEA CE, STM32CubeIDE, Git, GDB, Maven, Jacoco, AWS

Courses: Machine Learning, Intro Artificial Intelligence, Engineering Cloud Software Systems, Engineering of Software Subsystems, Software Design for Computing Systems, Web & Mobile I, Web Engineering